

Chapter 3 Elementary Number Theory

Fundamental

1. Natural Number, $\mathbb{N}$:

- **Definition:** Natural numbers are the set of positive integers used for counting and ordering. Typically, this set includes all whole numbers starting from 1 and increasing indefinitely $\{1, 2, 3, \dots\}$.

- **Example:** 1, 2, 3, ...

2. Integer, $\mathbb{Z}$:

- **Definition:** Integers are the set of whole numbers that can be positive, negative, or zero. This set includes all natural numbers, their negative counterparts, and zero.

- **Example:** ..., -3, -2, -1, 0, 1, 2, 3, ...

3. Rational Number, $\mathbb{Q}$:

- **Definition:** Rational numbers are any numbers that can be expressed as the quotient or fraction of two integers, where the numerator is an integer and the denominator is a non-zero integer. The set of rational numbers includes all integers and fractions. More formally, if r is a real number, then

$$r \text{ is rational} \Leftrightarrow \exists \text{ integers } a \text{ and } b \text{ such that } r = \frac{a}{b} \text{ and } b \neq 0.$$

- **Example:** $\frac{1}{2}$, -4, $3.75 = \frac{15}{4}$

4. Irrational Number:

- **Definition:** Irrational numbers are numbers that cannot be expressed as the quotient of two integers. They have non-repeating, non-terminating decimal expansions.

- **Example:** π , e , $\sqrt{2}$

5. Real Number, $\mathbb{R}$:

- **Definition:** Real numbers are all the numbers that can be found on the number line. This includes both rational and irrational numbers, encompassing all finite and infinite decimal expansions.

- **Example:** -2, 0, 1.5, π , e , $\sqrt{2}$

In summary:

- **Natural numbers** are positive counting numbers.
- **Integers** are whole numbers that can be positive, negative, or zero.
- **Rational numbers** are fractions or quotients of integers.
- **Irrational numbers** cannot be written as fractions and have non-repeating, non-terminating decimals.
- **Real numbers** include both rational and irrational numbers, covering all possible values on the number line

Even

An integer is called **even** provided it is **divisible** by two.

Example: ..., -4, -2, 0, 2, 4, ...

Divisible

Let a and b be integers. We say that a is **divisible** by b provided there is an integer c such that $bc = a$. We also say b divides a , or b is a factor of a , or b is a divisor of a . The notation for this is $b \mid a$.

Example 1

12 is divisible by 4. $4 \times 3 = 12$.

4 divides 12. 4 is a factor of 12. 4 is a divisor of 12.

$4 \mid 12$.

12 is not divisible by 5 because there is no integer x for which $5x = 12$; thus $5 \mid 12$ is false.

The number 12 is even because $2 \mid 12$, and we know $2 \mid 12$ because $2 \times 6 = 12$.

On the other hand, 13 is not even, because 13 is not divisible by 2; there is no integer x for which $2x = 13$.

Odd

An integer a is called **odd** provided there is an integer x such that $a = 2x + 1$.

Example: ..., -3, -1, 1, 3, ...

Thus 13 is odd because $13 = 2 \times 6 + 1$.

Prime

An integer p is called **prime** provided that $p > 1$ and the only positive divisors of p are 1 and p .

Example: 2, 3, 5, 7, 11, 13, 17, 19, 23, 31, ...

p_1, p_2, p_3

$p_{100} = ?$

$p_1 = ?$

For example, 11 is prime because it satisfies both conditions in the definition: First, 11 is greater than 1, and second, the only positive divisors of 11 are 1 and 11.

However, 12 is not prime because it has a positive divisor other than 1 and itself; for example, $3 \mid 12$, $3 \neq 1$, and $3 \neq 12$.

Is 1 a prime? No. To see why, take $p = 1$ and see if p satisfies the definition of primality.

There are two conditions: First we must have $p > 1$, and second, the only positive divisors of p are 1 and p . The second condition is satisfied: the only divisors of 1 are 1 and itself. However, $p = 1$ does not satisfy the first condition because $1 > 1$ is false.

Therefore, 1 is not a prime.

$$12 = 1 \times 2 \times 2 \times 3$$

$$12 = 1 \times 1 \times 2 \times 2 \times 3$$

many way
to represent 12

$$\begin{array}{r} 3 \\ 4 \overline{)12} \\ \underline{12} \\ 0 \end{array}$$

integer solution

$$\begin{array}{r} 2 \\ 5 \overline{)12} \\ \underline{10} \\ 2 \end{array}$$

$$\begin{array}{r} 6 \\ 2 \overline{)12} \\ \underline{12} \\ 0 \end{array}$$

only one way

Every positive integer $n > 1$ can be written uniquely as $p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s}$, where $p_1 < p_2 < \cdots < p_s$ are distinct primes that divide n and the k 's are positive integers giving the number of times each prime occurs as a factor of n .

Example 2

(a) $9 = 3 \cdot 3 = 3^2$

$$\begin{array}{r} 3 \overline{)9} \\ 3 \end{array}$$

$$\begin{array}{r} 2 \overline{)24} \\ 2 \overline{)12} \\ 2 \overline{)6} \\ 3 \end{array}$$

(b) $24 = 8 \cdot 3 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$

(c) $30 = 2 \cdot 3 \cdot 5$

$$\begin{array}{r} 2 \overline{)30} \\ 3 \overline{)15} \\ 5 \end{array}$$

Composite

A positive integer a is called **composite** provided there is an integer b such that $1 < b < a$ and $b \mid a$.

For example, the number 25 is composite because it satisfies the condition of the definition: There is a number b with $1 < b < 25$ and $b \mid 25$; indeed, $b = 5$ is the only such number. $25 = 5 \times 5$.

Prime numbers are not composite. If p is prime, then, by definition, there can be no divisor of p between 1 and p .

 $1 < b < p$, no such b

Furthermore, the number 1 is not composite. (Clearly, there is no number b with $1 < b < 1$.) Poor number 1! It is neither prime nor composite! (There is, however, a special term that is applied to the number 1 - the number 1 is called a unit.)

Dividing*unit identity.*

$1x = x$

$2 \times 2^{-1} = 1$

Division

Let $a, b \in \mathbb{Z}$ with $b > 0$. There exist integers q and r such that

$$a = qb + r \text{ and } 0 \leq r < b$$

$$\begin{array}{r} q \leftarrow \\ b \overline{)a} \\ c \\ \hline r \leftarrow \end{array}$$

Moreover, there is only one such pair of integers (q, r) that satisfies these conditions.

The integer q is called the **quotient** and the integer r is called the **remainder**.

Example 3

Let $a = 23$ and $b = 10$. Then the quotient is $q = 2$ and the remainder $r = 3$ because

$$23 = 2 \times 10 + 3 \text{ and } 0 \leq 3 < 10$$

$$a = qb + r \quad 0 \leq r < b$$

$$\begin{array}{r} 2 \leftarrow q \\ 10 \overline{)23} \\ 20 \\ \hline 3 \leftarrow r \end{array}$$

Example 4

Let $a = -37$ and $b = 5$. Then $q = -8$ and $r = 3$ because

$$-37 = -8 \times 5 + 3 \text{ and } 0 \leq 3 < 5$$

$$a = qb + r \quad 0 \leq r < b$$

$$\begin{array}{r} -8 \leftarrow q \\ 5 \overline{)-37} \\ -40 \\ \hline 3 \leftarrow r \end{array}$$

mod ?

Congruence modulo n

Let n be a positive integer. We say that integers x and y are **congruent modulo n** , and we write

$$x \equiv y \pmod{n}$$

provided $n \mid (x - y)$.

$$\begin{aligned} x - y &= kn \\ x &= kn + y \end{aligned}$$

In other words, $x \equiv y \pmod{n}$ if and only if x and y differ by a multiple of n .

Example 5

$$3 \equiv 13 \pmod{5}$$

because $3 - 13 = -10$ is a multiple of 5.

$$4 \equiv 4 \pmod{5}$$

because $4 - 4 = 0$ is a multiple of 5.

$$16 \not\equiv 3 \pmod{5}$$

because $16 - 3 = 13$ is not a multiple of 5.

$$-10 = (-2)5$$

$$0 = (0)5$$

$$13 = 2 \cdot 5 + 3$$

The next case is when $n = 2$. Two numbers are congruent mod 2 provided their difference is divisible by 2 (i.e., they differ by an even number). For example,

$$3 \equiv 15 \pmod{2}, \quad 0 \equiv -14 \pmod{2}, \quad \text{and} \quad 3 \equiv 3 \pmod{2}.$$

However,

$$\begin{aligned} 3 - 15 &= -12 = (-6)2 & 0 - (-14) &= 14 = (7)2 & 3 - 3 &= 0 = (0)2 \\ 3 - 12 &= -9 \neq k2 & 3 &\not\equiv 12 \pmod{2} & \text{and} & -1 \not\equiv 0 \pmod{2}. & (-1) - 0 &= -1 \neq k2. \end{aligned}$$

Please notice that two numbers are congruent mod 2 iff they are both even or both odd

Every integer is either even or odd, but not both.

Two integers are congruent modulo 2 if and only if they are both even or both odd.

Div and Mod

Let $a, b \in \mathbb{Z}$ with $b > 0$. There exists a unique pair of numbers q and r with $a = qb + r$ and $0 \leq r < b$. We define the operations **div** and **mod** by

$$a \text{ div } b = q \quad \text{and} \quad a \text{ mod } b = r.$$

$$a = qb + r, \quad 0 \leq r < b$$

$$\begin{array}{r} q \\ b \overline{) a} \\ \underline{c} \end{array}$$

Example 6

These calculations illustrate the div and mod operations.

$$\begin{array}{r} 3 \overline{) 11} \\ \underline{9} \\ 2 \end{array} \quad \begin{aligned} 11 \text{ div } 3 &= 3 \\ 11 \text{ mod } 3 &= 2 \end{aligned}$$

$$\begin{aligned} 23 \text{ div } 10 &= 2 & 23 \text{ mod } 10 &= 3 \\ -37 \text{ div } 5 &= -8 & -37 \text{ mod } 5 &= 3 \end{aligned}$$

$$\begin{array}{r} -8 \\ 5 \overline{) -37} \\ \underline{-40} \\ 3 \end{array} \quad \begin{aligned} -8 &= -37 \text{ div } 5 \\ 3 &= -37 \text{ mod } 5 \end{aligned}$$

$$\begin{array}{r} 2 \\ 10 \overline{) 23} \\ \underline{20} \\ 3 \end{array} \quad \begin{aligned} 2 &= 23 \text{ div } 10 \\ 3 &= 23 \text{ mod } 10 \end{aligned}$$

The meaning of $a \equiv b \pmod{n}$ is that $a - b$ is a multiple of n . We have $53 \equiv 23 \pmod{10}$ because $53 - 23 = 30$, a multiple of 10. $30 = (3) \cdot 10$ ✓

In the new meaning of this section, mod is a binary operation. For example,

$$53 \bmod 10 = 3$$

$$\begin{array}{r} 5 \\ 10 \overline{) 53} \\ \underline{50} \\ 3 \end{array}$$

$53 \bmod 10$

In this context, mod means "divide and take the remainder."

What is the connection between these two meanings of the word mod? We have the following result.

Let $a, b, n \in \mathbb{Z}$ with $n > 0$. Then

$$a \equiv b \pmod{n} \Leftrightarrow a \bmod n = b \bmod n$$

$$a - b = kn$$

Greatest Common Divisor

Common divisor

Let $a, b \in \mathbb{Z}$. We call an integer d a **common divisor** of a and b provided $d \mid a$ and $d \mid b$.

For example, the common divisors of 30 and 24 are $-6, -3, -2, -1, 1, 2, 3$, and 6 .

Greatest common divisor

$$\text{GCD}(30, 24) = 6$$

Let $a, b \in \mathbb{Z}$. We call an integer d the **greatest common divisor** of a and b provided

(1) d is a common divisor of a and b and

(2) if e is a common divisor of a and b , then $e \leq d$.

The greatest common divisor of a and b is denoted $\text{gcd}(a, b)$.

For example, the greatest common divisor of 30 and 24 is 6, and we write $\text{gcd}(30, 24) = 6$. Also $\text{gcd}(-30, -24) = 6$.

If d is $\text{GCD}(a, b)$, then

(a) $d = sa + tb$ for some integers s and t . (These are not necessarily positive.)

(b) If c is any other common divisor of a and b , then $c \mid d$.

Example 7

(a) The common divisors of 12 and 30 are 1, 2, 3, and 6, so that

$$\text{GCD}(12, 30) = 6 \text{ and } 6 = 1 \cdot 30 + (-2) \cdot 12$$

$$a \quad b$$

$$6 = (-2)12 + (1)30$$

$$d = sa + tb$$

(b) It is clear that $\text{GCD}(17, 95) = 1$ since 17 is prime and $17 \nmid 95$, and the reader may verify that $1 = 28 \cdot 17 + (-5) \cdot 95$.

$$d = 28a + (-5)b$$

If $\text{GCD}(a, b) = 1$, as in Example 7(b), we say a and b are **relatively prime**.

Euclidean algorithm *Exam*

One remaining question is that of how to compute the GCD conveniently in general.

We now present a procedure, called the Euclidean algorithm, for finding $\text{GCD}(a, b)$. Suppose that $a > b > 0$ (otherwise interchange a and b). Then by Theorem 1, we may write

$$a = k_1 b + r_1, \text{ where } k_1 \text{ is in } \mathbb{Z}^+ \text{ and } 0 \leq r_1 < b \quad (1)$$

if n divides a and b , then it must divide r_1 , since $r_1 = a - k_1 b$. Similarly, if n divides b and r_1 , then it must divide a . We see that the common divisors of a and b are the same as the common divisors of b and r_1 , so $\text{GCD}(a, b) = \text{GCD}(b, r_1)$.

$$\begin{array}{lll} \text{divide } b \text{ by } r_1: & b = k_2 r_1 + r_2 & 0 \leq r_2 < r_1 \\ \text{divide } r_1 \text{ by } r_2: & r_1 = k_3 r_2 + r_3 & 0 \leq r_3 < r_2 \\ \text{divide } r_2 \text{ by } r_3: & r_2 = k_4 r_3 + r_4 & 0 \leq r_4 < r_3 \\ \vdots & \vdots & \vdots \\ \text{divide } r_{n-2} \text{ by } r_{n-1}: & r_{n-2} = k_n r_{n-1} + r_n & 0 \leq r_n < r_{n-1} \\ \text{divide } r_{n-1} \text{ by } r_n: & r_{n-1} = k_{n+1} r_n + r_{n+1} & 0 \leq r_{n+1} < r_n \end{array} \quad (2)$$

Since $a > b > r_1 > r_2 > r_3 > r_4 > \dots$, the remainder will eventually become zero, so at some point we have $r_{n+1} = 0$.

We now show that $r_n = \text{GCD}(a, b)$. We saw previously that

$$\text{GCD}(a, b) = \text{GCD}(b, r_1)$$

Repeating this argument with b and r_1 , we see that

$$\text{GCD}(b, r_1) = \text{GCD}(r_1, r_2)$$

Upon continuing, we have

$$\text{GCD}(a, b) = \text{GCD}(b, r_1) = \text{GCD}(r_1, r_2) = \dots = \text{GCD}(r_{n-1}, r_n)$$

Since $r_{n-1} = k_{n+1} r_n$, we see that $\text{GCD}(r_{n-1}, r_n) = r_n$. Hence $r_n = \text{GCD}(a, b)$.

Example 8

Compute GCD (273,98). Let a be 273 and b be 98. Using the Euclidean algorithm,

divide 273 by 98: $273 = 2 \cdot 98 + 77$

divide 98 by 77: $98 = 1 \cdot 77 + 21$

divide 77 by 21: $77 = 3 \cdot 21 + 14$

divide 21 by 14: $21 = 1 \cdot 14 + 7$

divide 14 by 7: $14 = 2 \cdot 7 + 0$

so $\text{GCD}(273,98) = 7$, the last of the nonzero divisors.

$$\begin{array}{r}
 98 \overline{)273} \\
 \underline{196} \\
 77 \\
 \\
 77 \overline{)98} \\
 \underline{77} \\
 21 \\
 \\
 21 \overline{)77} \\
 \underline{63} \\
 14 \\
 \\
 14 \overline{)21} \\
 \underline{14} \\
 7 \\
 \\
 7 \overline{)14} \\
 \underline{14} \\
 0
 \end{array}$$

$273 = 2 \cdot 98 + 77$	$77 = 273 - 2 \cdot 98$
$98 = 1 \cdot 77 + 21$	$21 = 98 - 1 \cdot 77$
$77 = 3 \cdot 21 + 14$	$14 = 77 - 3 \cdot 21$
$21 = 1 \cdot 14 + 7$	$7 = 21 - 1 \cdot 14$
$14 = 2 \cdot 7 + 0$	

$$\begin{array}{cc}
 \text{GCD}(273, 98) = 7 & d = 7 \\
 \begin{array}{ccc} a & b & d \end{array} &
 \end{array}$$

$$d = sa + tb, \quad s = ?, \quad t = ?$$

We observed that if $d = \text{GCD}(a, b)$, we can find integers s and t such that $d = sa + tb$. The integers s and t can be found as follows. Solve the next-to-last equation in (2) for r_n :

$$r_n = r_{n-2} - k_n r_{n-1} \quad (3)$$

Now solve the second-to-last equation in (2), $r_{n-3} = k_{n-1} r_{n-2} + r_{n-1}$ for r_{n-1} :

$$r_{n-1} = r_{n-3} - k_{n-1} r_{n-2}$$

and substitute this expression in (3):

$$r_n = r_{n-2} - k_n [r_{n-3} - k_{n-1} r_{n-2}]$$

Continue to work up through the equations in (2) and (1), replacing r_i by an expression involving r_{i-1} and r_{i-2} , and finally arriving at an expression involving only a and b .

Example 9

(a) Let a be 273 and b be 98, as in Example 8. Then

$$\begin{aligned}
 d = \text{GCD}(273, 98) &= 7 = 21 - 1 \cdot 14 \\
 &= 21 - 1(77 - 3 \cdot 21) & 14 &= 77 - 3 \cdot 21 \\
 &= 4 \cdot 21 - 1 \cdot 77 \\
 &= 4(98 - 1 \cdot 77) - 1 \cdot 77 & 21 &= 98 - 1 \cdot 77 \\
 &= 4 \cdot 98 - 5 \cdot 77 \\
 &= 4 \cdot 98 - 5(273 - 2 \cdot 98) & 77 &= 273 - 2 \cdot 98 \\
 &= 14(98) - 5(273) = -5(273) + 14(98) \\
 &= s a + t b
 \end{aligned}$$

Hence $s = -5$ and $t = 14$. Note that the key is to carry out the arithmetic only partially.

$$\begin{aligned}
 d = 7 &= 21 - 1 \cdot 14 \\
 &= 21 - 1 \cdot (77 - 3 \cdot 21) \\
 &= 4 \cdot 21 - 1 \cdot 77 \\
 &= 4(98 - 1 \cdot 77) - 1 \cdot 77 \\
 &= 4 \cdot 98 - 5 \cdot 77 \\
 &= 4 \cdot 98 - 5(273 - 2 \cdot 98) \\
 &= 14 \cdot 98 - 5 \cdot 273 \\
 7 &= (-5)273 + (14)98 \quad (\text{check}) \\
 d &= s a + t b
 \end{aligned}$$

(b) Let $a = 108$ and $b = 60$. Then

$$\begin{aligned}
 \text{GCD}(108, 60) &= 12 = 60 - 1(48) \\
 &= 60 - 1[108 - 1(60)] & 48 &= 108 - 1 \cdot 60 \\
 &= 2(60) - 108
 \end{aligned}$$

$$\begin{array}{r}
 60 \overline{)108} \\
 \underline{60} \\
 48
 \end{array}
 \qquad
 \begin{array}{r}
 48 \overline{)60} \\
 \underline{48} \\
 12
 \end{array}$$

Hence $s = -1$ and $t = 2$.

$$\begin{aligned}
 108 &= 1 \cdot 60 + 48 & 48 &= 108 - 1 \cdot 60 \\
 60 &= 1 \cdot 48 + 12 & 12 &= 60 - 1 \cdot 48 \\
 48 &= 4 \cdot 12 + 0 \\
 \text{GCD}(60, 48) &= d = 12. \\
 d = 12 &= 60 - 1 \cdot 48 \\
 &= 60 - 1(108 - 1 \cdot 60) \\
 &= 2 \cdot 60 + (-1)108 \\
 12 &= (-1)108 + (2)60 \quad (\text{check}) \\
 d &= s a + t b
 \end{aligned}$$

Example 10Calculating $\gcd(431, 29)$ by Euclid's Algorithm: $a \quad b$

$$431 = 14 \times 29 + 25$$

$$29 = 1 \times 25 + 4$$

$$25 = 6 \times 4 + 1$$

$$4 = 4 \times 1 + 0$$

$\uparrow$ 431 and 29 are relatively prime.
 $\gcd(431, 29) = 1$

In all of these equations (except the last), we solve for the remainder (that is, we put the remainders on the left).

$$25 = 431 - 14 \times 29$$

$$4 = 29 - 1 \times 25$$

$$1 = 25 - 6 \times 4$$

$\uparrow$

Now we work from the bottom up. Notice that the last equation has 1 in the form $25x + 4y$. We substitute for 4 using the previous equation:

$$1 = 25 - 6 \times 4$$

$$= 25 - 6 \times (29 - 1 \times 25)$$

$$= -6 \times 29 + 7 \times 25$$

Now we use $25 = 431 - 14 \times 29$ to replace the 25 in $1 = -6 \times 29 + 7 \times 25$:

$$1 = -6 \times 29 + 7 \times 25$$

$$= -6 \times 29 + 7 \times (431 - 14 \times 29)$$

$$= 7 \times 431 + [-6 + 7 \times (-14)]29$$

$$1 = 7 \times 431 + (-104) \times 29$$

$$d = xa + yb$$

$$1 = xa + yb$$

This is how we found $x = 7$ and $y = -104$ to get $431x + 29y = \gcd(431, 29) = 1$.

$$431x + 29y = \gcd(431, 29) = 1$$

$$431x + 29y = 1$$

$$(431x + 29y) \bmod 431 = 1 \bmod 431$$

$$29y \bmod 431 = 1 \bmod 431$$

$$29(-104) \bmod 431 = 1 \bmod 431$$

Remark.

$$1 = 7 \times 431 + (-104) \times 29$$

$$1 = x \cdot 431 + y \cdot 29$$

Pairs of numbers whose greatest common divisor is 1 have a special name.

Let a and b be integers. We call a and b **relatively prime** provided $\gcd(a, b) = 1$.

In other words, integers are relatively prime provided the only divisors they have in common are 1 and -1 .

Let a and b be integers. There exist integers x and y such that $ax + by = 1$ if and only if a and b are relatively prime.

Least Common Multiple

If a, b , and k are in $\mathbb{Z}^+$, and $a|k, b|k$, we say k is a common multiple of a and b . The smallest such k , call it c , is called the least common multiple, or LCM, of a and b , and we write $c = \text{LCM}(a, b)$. The following result shows that we can obtain the least common multiple from the greatest common divisor, so we do not need a separate procedure for finding the least common multiple.

If a and b are two positive integers, then $\text{GCD}(a, b) \cdot \text{LCM}(a, b) = ab$.

Let $p_1, p_2, \dots, p_k$ be all the prime factors of either a or b . Then we can write

$$a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \text{ and } b = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$$

where some of the a_i and b_i may be zero. It then follows that

$$\text{GCD}(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_k^{\min(a_k, b_k)}$$

and

$$\text{LCM}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_k^{\max(a_k, b_k)}$$

} For understanding slow method.

Hence

$$\begin{aligned} \text{GCD}(a, b) \cdot \text{LCM}(a, b) &= p_1^{a_1+b_1} p_2^{a_2+b_2} \cdots p_k^{a_k+b_k} \\ &= (p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) \cdot (p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}) \\ &= ab \quad \checkmark \end{aligned}$$

$$\text{LCM}(a, b) = \frac{ab}{\text{GCD}(a, b)} \quad *$$

Example 11

Let $a = 540$ and $b = 504$. Factoring a and b into primes, we obtain

$$a = 540 = 2^2 \cdot 3^3 \cdot 5 \text{ and } b = 504 = 2^3 \cdot 3^2 \cdot 7$$

Thus all the prime numbers that are factors of either a or b are $p_1 = 2, p_2 = 3, p_3 = 5$, and $p_4 = 7$. Then $a = 2^2 \cdot 3^3 \cdot 5^1 \cdot 7^0$ and $b = 2^3 \cdot 3^2 \cdot 5^0 \cdot 7^1$. We then have

$$\begin{aligned}\text{GCD}(540, 504) &= 2^{\min(2,3)} \cdot 3^{\min(3,2)} \cdot 5^{\min(1,0)} \cdot 7^{\min(0,1)} \\ &= 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^0 \\ &= 2^2 \cdot 3^2 \text{ or } 36\end{aligned}$$

Also,

$$\begin{aligned}\text{LCM}(540, 504) &= 2^{\max(2,3)} \cdot 3^{\max(3,2)} \cdot 5^{\max(1,0)} \cdot 7^{\max(0,1)} \\ &= 2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \text{ or } 7560\end{aligned}$$

Then

$$\text{GCD}(540, 504) \cdot \text{LCM}(540, 504) = 36 \cdot 7560 = 272,160 = 540 \cdot 504.$$

As a verification, we can also compute GCD(540, 504) by the Euclidean algorithm and obtain the same result.

$$\text{LCM}(540, 504) = \frac{540 \times 504}{\text{GCD}(540, 504)} = \frac{540 \times 504}{36} = 7560 \quad \text{Fastest,}$$

Use Euclidean Algorithm (Exam)

$$540 = 1 \cdot 504 + 36$$

$$504 = 14 \cdot 36 + 0$$

$$\begin{aligned}\text{LCM}(540, 504) &= \frac{540 \times 504}{\text{GCD}(540, 504)} \\ &= \frac{540 \times 504}{36} \\ &= 7560\end{aligned}$$

$$\begin{array}{r} 1 \\ 504 \overline{) 540} \\ \underline{404} \\ 36 \end{array}$$

$$\begin{array}{r} 14 \\ 36 \overline{) 504} \\ \underline{504} \\ 0 \end{array}$$

Modular Arithmetic

$$\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\} \quad \mathbb{Z}_3 = \{0, 1, 2\}$$

that is, $\mathbb{Z}_n$ contains all natural numbers from 0 to $n-1$ inclusive.

How are the modular operations defined? We begin with $\oplus$ and $\otimes$.

Modular addition, multiplication

Let n be a positive integer. Let $a, b \in \mathbb{Z}_n$. We define

$$\begin{array}{r} n \overline{)ab} \\ \underline{} \\ (ab) \bmod n \end{array}$$

$$a \oplus b = (a + b) \bmod n \text{ and } a \otimes b = (ab) \bmod n$$

$$\begin{array}{r} n \overline{)a+b} \\ \underline{} \\ (a+b) \bmod n \end{array}$$

The operations on the left are operations defined for $\mathbb{Z}_n$. The operations on the right are ordinary integer operations.

Example 12 $\mathbb{Z}_{10} = \{0, 1, 2, \dots, 9\}$

Let $n = 10$. In $\mathbb{Z}_{10}$, we have the following:

$$\begin{array}{ll} 5 \oplus 5 = 0 & 9 \oplus 8 = 7 \\ 5 \otimes 5 = 5 & 9 \otimes 8 = 2 \end{array}$$

$$\begin{array}{r} 10 \overline{)10} \\ \underline{10} \\ 0 \in \mathbb{Z}_{10} \end{array}$$

Notice that the symbols $\oplus$ and $\otimes$ depend on the context.

If we are working in $\mathbb{Z}_{10}$, then $5 \oplus 5 = 0$, but if we are working in $\mathbb{Z}_9$, then $5 \oplus 5 = 1$.

$\begin{array}{l} \mathbb{Z}_{10} \\ 5 \oplus 5 = (5+5) \bmod 10 \\ = 10 \bmod 10 \\ = 0 \\ 9 \oplus 8 = (9+8) \bmod 10 \\ = 17 \bmod 10 \\ = 7 \end{array}$	$\begin{array}{l} \mathbb{Z}_9 \\ 5 \oplus 5 = (5+5) \bmod 9 \\ = 10 \bmod 9 \\ = 1 \end{array}$
--	--

Notice that if $a, b \in \mathbb{Z}_n$, the results of the operations $a \oplus b$ and $a \otimes b$ are always defined and are elements of $\mathbb{Z}_n$.

Let $a, b \in \mathbb{Z}_n$. Then $a \oplus b \in \mathbb{Z}_n$ and $a \otimes b \in \mathbb{Z}_n$. (Closure.)

The operations $\oplus$ and $\otimes$ exhibit the usual algebraic properties:

Let n be an integer with $n \geq 2$.

- For all $a, b \in \mathbb{Z}_n$, $a \oplus b = b \oplus a$ and $a \otimes b = b \otimes a$. (Commutative.)
- For all $a, b, c \in \mathbb{Z}_n$, $a \oplus (b \oplus c) = (a \oplus b) \oplus c$, and $a \otimes (b \otimes c) = (a \otimes b) \otimes c$. (Associative.)

- For all $a \in \mathbb{Z}_n$, $a \oplus 0 = a$, $a \otimes 1 = a$, and $a \otimes 0 = 0$. (Identity elements: 0 for addition and 1 for multiplication. Note that 0 is not an identity element for multiplication.)
- For all $a, b, c \in \mathbb{Z}_n$, $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$. (Distributive.)

Modular Subtraction

Let n be a positive integer and let $a, b \in \mathbb{Z}_n$. Then $a \ominus b = (a - b) \bmod n$.

Example 13

$$\mathbb{Z}_{10} = \{0, 1, 2, \dots, 9\}$$

Let $n = 10$.

$$5 \ominus 3 = (5 - 3) \bmod 10 = 2 \bmod 10 = 2.$$

$$3 \ominus 5 = (3 - 5) \bmod 10 = -2 \bmod 10 = 8.$$

$$\begin{aligned} 3 \ominus 5 &= (3 - 5) \bmod 10 \\ &= -2 \bmod 10 \\ &= 8 \end{aligned}$$

$$\begin{array}{r} -1 \\ 10 \overline{) -2} \\ \underline{-10} \\ 8 \end{array}$$

Modular reciprocal

Let n be a positive integer and let $a \in \mathbb{Z}_n$. A **reciprocal** of a is an element $b \in \mathbb{Z}_n$ such that $a \otimes b = 1$. An element of $\mathbb{Z}_n$ that has a reciprocal is called invertible.

Let's investigate reciprocals in $\mathbb{Z}_{10}$. Here is the multiplication table for $\mathbb{Z}_{10}$:

$\otimes$	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	5	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

$$a \otimes a^{-1} = 1$$

$$1 \otimes 1 = 1$$

$$1^{-1} = 1$$

$$3 \otimes 7 = 1$$

$$3^{-1} = 7$$

$$7^{-1} = 3$$

$$9 \otimes 9 = 1$$

$$9^{-1} = 9$$

$$\{1, 3, 7, 9\}$$

Several comments are in order.

- Element 0 does not have a reciprocal; this is not surprising.
- Elements 2, 4, 5, 6, and 8 do not have reciprocals. This explains why our attempts to divide by 2 were strange.
- Elements 1, 3, 7, and 9 are invertible (have reciprocals). Furthermore, they have only one reciprocal each.
- Notice the elements of $\mathbb{Z}_{10}$ that have reciprocals are precisely those integers in $\mathbb{Z}_{10}$ that are relatively prime to 10.
- The reciprocal of 3 is 7, and the reciprocal of 7 is 3; both 1 and 9 are their own reciprocals.

$$\{1, 3, 7, 9\} \text{ are relatively prime to } 10$$

$$\begin{array}{ll} 3^{-1} = 7 & 1^{-1} = 1 \\ 7^{-1} = 3 & 9^{-1} = 9 \end{array}$$

$$\text{In } \mathbb{Z}_{10}$$

Let n be a positive integer and let $a \in \mathbb{Z}_n$. If a has a reciprocal in $\mathbb{Z}_n$, then it has only one reciprocal.

Modular division

Let n be a positive integer and let b be an invertible element of $\mathbb{Z}_n$. Let $a \in \mathbb{Z}_n$ be arbitrary.

Then $a \oslash b$ is defined to be $a \otimes b^{-1}$.

Example 14

In $\mathbb{Z}_{10}$, calculate $2 \oslash 7$.

Note that $7^{-1} = 3$.

So $2 \oslash 7 = 2 \otimes 7^{-1} = 2 \otimes 3 = 6$.

$$\begin{aligned} 2 \oslash 7 &= 2 \otimes 7^{-1} \\ &= 2 \otimes 3 \\ &= (2 \times 3) \bmod 10 \\ &= 6 \bmod 10 \\ 2 \oslash 7 &= 6 \end{aligned}$$

We still have some work to do. We need to address the following issues:

- In $\mathbb{Z}_n$, which elements are invertible?
- In $\mathbb{Z}_n$, given that a is invertible, how do we calculate a^{-1} ?

We solved these problems for $\mathbb{Z}_{10}$ by writing out the entire $\otimes$ table for $\mathbb{Z}_{10}$. We would not want to do that for $\mathbb{Z}_{1000}$!

We noticed that the only invertible elements in $\mathbb{Z}_{10}$ are 1, 3, 7, and 9 - precisely those elements relatively prime to 10. Does this pattern continue? Let's examine $\mathbb{Z}_9$. Here is the $\otimes$ table for $\mathbb{Z}_9$:

$\otimes$	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8
2	0	2	4	6	8	1	3	5	7
3	0	3	6	0	3	6	0	3	6
4	0	4	8	3	7	2	6	1	5
5	0	5	1	6	2	7	3	8	4
6	0	6	3	0	6	3	0	6	3
7	0	7	5	3	1	8	6	4	2
8	0	8	7	6	5	4	3	2	1

$$1^{-1} = 1$$

$$2^{-1} = 5, 5^{-1} = 2$$

$$4^{-1} = 7, 7^{-1} = 4$$

$$8^{-1} = 8$$

$$1 \otimes 1 = 1$$

$$2 \otimes 5 = 1$$

$$4 \otimes 7 = 1$$

$$8 \otimes 8 = 1$$

$$3 \otimes 5 = 1$$

$$= 3 \otimes 5^{-1}$$

$$= 3 \otimes 2$$

$$= 6$$

The invertible elements of $\mathbb{Z}_9$ are 1, 2, 4, 5, 7, and 8 (these are all relatively prime to 9), and the noninvertible elements are 0, 3, and 6 (none of these are relatively prime to 9).

This suggests the following.

what if n is a large number.

Invertible elements of $\mathbb{Z}_n$

Let n be a positive integer and let $a \in \mathbb{Z}_n$.

Then a is invertible if and only if a and n are relatively prime.

We now know that the invertible elements of $\mathbb{Z}_n$ are exactly those that are relatively prime to n .

Let $a \in \mathbb{Z}_n$ and suppose $\gcd(a, n) = 1$.

Thus there are integers x and y so that $ax + ny = 1$.

To find the numbers x and y , we use back substitution in Euclid's algorithm.

Example 15

In $\mathbb{Z}_{431}$, find 29^{-1} .

$$29 \otimes y = 1 \quad 29y = 1 \pmod{431}$$

$$ax + by = d$$

Solution. In Example 10, we found integers x and y such that $431x + 29y = 1$, namely $x = 7$ and $y = -104$. Therefore $(-104 \cdot 29) \pmod{431} = 1$.

However, $-104 \notin \mathbb{Z}_{431}$. Instead we can take

$$z = -104 \pmod{431} = 327$$

Now $29 \otimes 327 = (29 \cdot 327) \pmod{431} = 9483 \pmod{431} = 1$. Therefore $29^{-1} = 327$.

$$\begin{array}{r} x \\ 431 \overline{) 431x} \\ \underline{431x} \\ 0 \end{array}$$

(E10)

$\mathbb{Z}_{431}$

$$(431x + 29y) \pmod{431} = 1 \pmod{431}$$

$$\underline{431x \pmod{431}} + 29y \pmod{431} = 1 \pmod{431}$$

$$+ 29y \pmod{431} = 1 \pmod{431}$$

$$29 \otimes 327 = 1$$

$$29^{-1} = 327$$

$$y = -104 \pmod{431}$$

$$y = 327$$

$$\begin{array}{r} -1 \\ 431 \overline{) -104} \\ \underline{-431} \\ 327 \end{array}$$

$$a \frac{b}{c}$$

Example 16

In $\mathbb{Z}_{431}$, calculate $30 \oslash 29$

Solution. In the previous example, we found $29^{-1} = 327$. Therefore

$$30 \oslash 29 = 30 \otimes 29^{-1} = 30 \otimes 327 = (30 \cdot 327) \bmod 431 = 9810 \bmod 431 = 328$$

$$30 \oslash 29 = 30 \otimes 29^{-1} = 30 \otimes 327 = (30 \cdot 327) \bmod 431 \\ = (9810) \bmod 431$$

$$30 \oslash 29 = 328$$

The Chinese Remainder Theorem

In this section, we investigate how to solve equations that involve modular equivalences.

Solving One Equation

We start with an easy example.

Example 17

Solve the equation

$$x \equiv 4 \pmod{11}$$

Solution.

$$x = 4 + 11k$$

This asks for all integers x such that $x - 4$ is a multiple of 11 (i.e., $x - 4 = 11k$ for some integer k).

We can rewrite this as $x = 4 + 11k$ where k can be any integer.

So the solutions are $\dots, -18, -7, 4, 15, 26, \dots$

$$\{ x = 4 + 11k \mid k \in \mathbb{Z} \}$$

Example 18

Solve the equation

$$3x \equiv 4 \pmod{11}$$

(43)

Suppose we had a solution x_0 to the equation $3x \equiv 4 \pmod{11}$.

Now consider the integer $x_1 = x_0 + 11$. If we substitute x_1 for x in Equation (43), we get

$$3x_1 = 3(x_0 + 11) = 3x_0 + 33 \equiv 3x_0 \equiv 4 \pmod{11}$$

so x_1 is also a solution.

$$3 \otimes x = 4 \\ 3^{-1} \otimes 3 \otimes x = 3^{-1} \otimes 4 \\ 1 \otimes x = 4 \otimes 4 \\ x = 5$$

$$3 \otimes 4 = 1 \\ 3^{-1} = 4$$

$$431 \overline{) 9810} \\ \underline{328}$$

$$x \equiv 5 \pmod{11}$$

$$\{5 + 11k \mid k \in \mathbb{Z}\}$$

$$\{x : x \equiv 5 \pmod{11}\}$$

Thus, if we add or subtract any multiple of 11 to a solution to Equation (43), we obtain another solution to Equation (43).

So if there is a solution, then there is a solution in $\{0, 1, 2, \dots, 10\} = \mathbb{Z}_{11}$.

Once we find all the solutions in $\mathbb{Z}_{11}$, we have found all solutions to the equation.

We seek a number $x \in \mathbb{Z}_{11}$ for which $3x \equiv 4 \pmod{11}$. But note that

$$3x \equiv 4 \pmod{11} \Leftrightarrow (3x) \bmod 11 = 4 \Leftrightarrow 3 \otimes x = 4$$

where $\otimes$ is modular multiplication in $\mathbb{Z}_{11}$

We multiply both sides of $3 \otimes x = 4$ by 3^{-1} .

$3^{-1} = 4$, and so

$$3 \otimes x = 4 \Rightarrow 3^{-1} \otimes 3 \otimes x = 3^{-1} \otimes 4 \Rightarrow 4 \otimes 3 \otimes x = 4 \otimes 4 \Rightarrow 1 \otimes x = 5 \Rightarrow x = 5$$

(because $12 \bmod 11 = 1$ and $16 \bmod 11 = 5$).

Let's check this answer in Equation (43). We substitute $x = 5$ and calculate

$$3x = 15 \equiv 4 \pmod{11}$$

and so 5 is a solution.

The full set of solutions is $\{5 + 11k : k \in \mathbb{Z}\} = \{\dots, -17, -6, 5, 16, 27, \dots\}$.

$\{x : x \equiv 5 \pmod{11}\}$.

This completes the solution to Example 18.

$$3x \equiv 4 \pmod{11}$$

$$\mathbb{Z}_{11} = \{0, 1, 2, \dots, 10\}$$

$$3 \otimes x = 4$$

$$3^{-1} \otimes 3 \otimes x = 3^{-1} \otimes 4$$

$$1 \otimes x = 4 \otimes 4$$

$$x = 5$$

$$x \equiv 5 \pmod{11}$$

$$\{5 + 11k : k \in \mathbb{Z}\}$$

$$\{x : x \equiv 5 \pmod{11}\}$$

$$3 \otimes y = 1$$

$$3 \otimes 4 = 1$$

$$3^{-1} = 4$$

We summarize what we have learned in the following result.

Let $a, b, n \in \mathbb{Z}$ with $n > 0$. Suppose a and n are relatively prime and consider the equation

$$ax \equiv b \pmod{n}$$

The set of solutions to this equation is

$$\{x_0 + kn : k \in \mathbb{Z}\} = \{x : x \equiv x_0 \pmod{n}\}.$$

where $x_0 = a_0^{-1} \otimes b_0$, $a_0 = a \bmod n$, $b_0 = b \bmod n$, and $\otimes$ is modular multiplication in $\mathbb{Z}_n$. The integer x_0 is the only solution to this equation in $\mathbb{Z}_n$.

Solving Two Equations

Now we solve a pair of congruence equations in different moduli. The type of problem we solve is

$$\begin{aligned} x &\equiv a \pmod{m} \text{ and} \\ x &\equiv b \pmod{n} \end{aligned}$$

Let's work out the solution to the following problem.

Example 19

Solve the pair of equations

$$\begin{aligned} x &\equiv 1 \pmod{7} \text{ and} \\ x &\equiv 4 \pmod{11} \end{aligned}$$

In other words, we want to find all integers x that satisfy both of these equations.

Let's begin with the first equation. Since $x \equiv 1 \pmod{7}$, we can write

$$x = 1 + 7k$$

for some integer k . We can substitute $1 + 7k$ for x in the second equation:

$x \equiv 4 \pmod{11}$. This gives

$$1 + 7k \equiv 4 \pmod{11} \Rightarrow 7k \equiv 3 \pmod{11}$$

The problem now reduces to a single equation in k . To solve this equation, we need to $\otimes$ both sides by 7^{-1} working in $\mathbb{Z}_{11}$. In $\mathbb{Z}_{11}$ we find that $7^{-1} = 8$. We calculate, in $\mathbb{Z}_{11}$,

$$7 \otimes k = 3 \Rightarrow 8 \otimes 7 \otimes k = 8 \otimes 3 \Rightarrow k = 2$$

We can check that $7^{-1} = 8$ by calculating $7 \otimes 8 = (7 \cdot 8) \bmod 11 = 56 \bmod 11 = 1$

Furthermore, if we increase or decrease $k = 2$ by any multiple of 11, we again have a solution to $1 + 7k \equiv 4 \pmod{11}$.

We are nearly finished. Let's write down what we have. We know that we want all values of x with

$$x = 1 + 7k$$

and k can be any integer of the form

$$k = 2 + 11j$$

where j is any integer. Combining these two, we have

$$x = 1 + 7k = 1 + 7(2 + 11j) = 15 + 77j \quad (\forall j \in \mathbb{Z})$$

In other words, the solution set to the equations in Example 19 is

$$\{x \in \mathbb{Z} : x \equiv 15 \pmod{77}\}.$$

To check that this is correct, notice that

$$15 \equiv 1 \pmod{7} \quad \text{and} \quad 15 \equiv 4 \pmod{11}$$

Furthermore, if x is increased or decreased by any multiple of 77, both equations remain valid because 77 is a multiple of both 7 and 11.

Handwritten derivation showing the solution to the system of congruences:

$$\begin{cases} x \equiv 1 \pmod{7} & (1) \\ x \equiv 4 \pmod{11} & (2) \end{cases}$$

From (1): $x = 1 + 7k$

Substitute into (2): $1 + 7k \equiv 4 \pmod{11}$

$7k \equiv 3 \pmod{11}$

Find the inverse of 7 modulo 11: $7 \otimes y = 1 \pmod{11}$

$7 \otimes 8 = 1 \pmod{11}$

$7^{-1} = 8$

$1 \otimes k = 8 \otimes 3$

$k \equiv 24 \pmod{11}$

$k \equiv 2 \pmod{11}$

$k = 2 + 11j$

Substitute back into (1): $x = 1 + 7k$

$x = 1 + 7(2 + 11j)$

$x = 15 + 77j$

Final solution set: $\{15 + 77j : j \in \mathbb{Z}\}$

Equation: $77 = 7 \times 11$

Final solution set: $\{x : x \equiv 15 \pmod{77}\}$

Chinese Remainder Theorem

Let a, b, m, n be integers with m and n positive and relatively prime. There is a unique integer x_0 with $0 \leq x_0 < mn$ that solves the pair of equations

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases} \text{ and}$$

$$\gcd(m, n) = 1$$

Furthermore, every solution to these equations differs from x_0 by a multiple of mn .

$$\{x_0 + mnj : j \in \mathbb{Z}\}$$

$$\{x : x \equiv x_0 \pmod{mn}\}$$

Example 20

Suppose we want to solve a system of three equations. For example, solve for all x :

$$\begin{cases} x \equiv 3 \pmod{9} \\ x \equiv 5 \pmod{10}, \text{ and} \\ x \equiv 2 \pmod{11} \end{cases}$$

$$\gcd(9, 10) = 1$$

$$\gcd(90, 11) = 1$$

$$\{x_0 + 9 \times 10 \times 11j : j \in \mathbb{Z}\}$$

$$\{x : x \equiv x_0 \pmod{9 \times 10 \times 11}\}$$

Solution: We can solve the first two equations by the usual method

$$\begin{cases} x \equiv 3 \pmod{9} \\ x \equiv 5 \pmod{10} \end{cases} \Rightarrow x \equiv 75 \pmod{90}$$

Now we combine this result with the last equation and solve again by the usual method.

$$\begin{cases} x \equiv 75 \pmod{90} \\ x \equiv 2 \pmod{11} \end{cases} \Rightarrow x \equiv 255 \pmod{990}$$

How?

$$x \equiv 3 \pmod{9}$$

$$x = 3 + 9j$$

$$x \equiv 5 \pmod{10}$$

$$3 + 9j \equiv 5 \pmod{10}$$

$$9j \equiv 2 \pmod{10}$$

$$\mathbb{Z}_{10}$$

$$9 \otimes y = 1$$

$$9 \otimes 9 = 1$$

$$9^{-1} = 9$$

$$9^{-1} \otimes 9 \otimes j = 9^{-1} \otimes 2$$

$$1 \otimes j = 9 \otimes 2$$

$$j = 8$$

$$j = 8 + 10k$$

$$x = 3 + 9j$$

$$x = 3 + 9(8 + 10k)$$

$$x = 75 + 90k$$

$$\begin{cases} x \equiv 75 \pmod{90} \\ x \equiv 2 \pmod{11} \end{cases}$$

$$\begin{cases} x \equiv 2 \pmod{11} \\ x \equiv 75 \pmod{90} \end{cases} \quad \gcd(11, 90) = 1$$

$$x \equiv 2 \pmod{11}$$

$$\bullet x = 2 + 11m$$

$$2 + 11m \equiv 75 \pmod{90} \quad \mathbb{Z}_{90}$$

$$11 \otimes m = 73$$

$$11^{-1} \otimes 11 \otimes m = 11^{-1} \otimes 73$$

$$1 \otimes m = 41 \otimes 73$$

$$11 \otimes y = 1$$

$$11 \otimes 41 = 1$$

$$11^{-1} = 41$$

$$m = 23$$

$$m = 23 + 90n$$

$$\bullet x = 2 + 11(23 + 90n)$$

$$x = 255 + 990n$$

$$\{255 + 990n : n \in \mathbb{Z}\}$$

$$\{x : x \equiv 255 \pmod{990}\}$$